Analyzing Student's Mental Health



Executive Summary

This project analyzes the psychological impact of studying abroad on international students. Using **PostgreSQL**, I queried and aggregated mental health survey data to uncover how the length of stay correlates with depression (PHQ), social connectedness (SCS), and acculturative stress (AS). The primary objective is to provide actionable data for university administrations to optimize the allocation of student wellness resources.

Does going to university in a different country affect your mental health? A Japanese international university surveyed its students in 2018 and published a study the following year that was approved by several ethical and regulatory boards.

The study found that international students have a higher risk of mental health difficulties than the general population, and that social connectedness (belonging to a social group) and acculturative stress (stress associated with joining a new culture) are predictive of depression.

Explore the `students` data using PostgreSQL to find out if you would come to a similar conclusion for international students and see if the length of stay is a contributing factor.

Here is a data description of the columns you may find helpful.

Field Name	Description
`inter_dom`	Types of students (international or domestic)
`japanese_cate`	Japanese language proficiency
`english_cate`	English language proficiency
`academic`	Current academic level (undergraduate or graduate)
`age`	Current age of student
`stay`	Current length of stay in years
`todep`	Total score of depression (PHQ-9 test)
`tosc`	Total score of social connectedness (SCS test)
`toas`	Total score of acculturative stress (ASISS test)

Are international students at the Japanese university at a higher risk of mental health difficulties (such as depression) compared to domestic students? Additionally, is the length of stay in the country associated with these mental health outcomes?

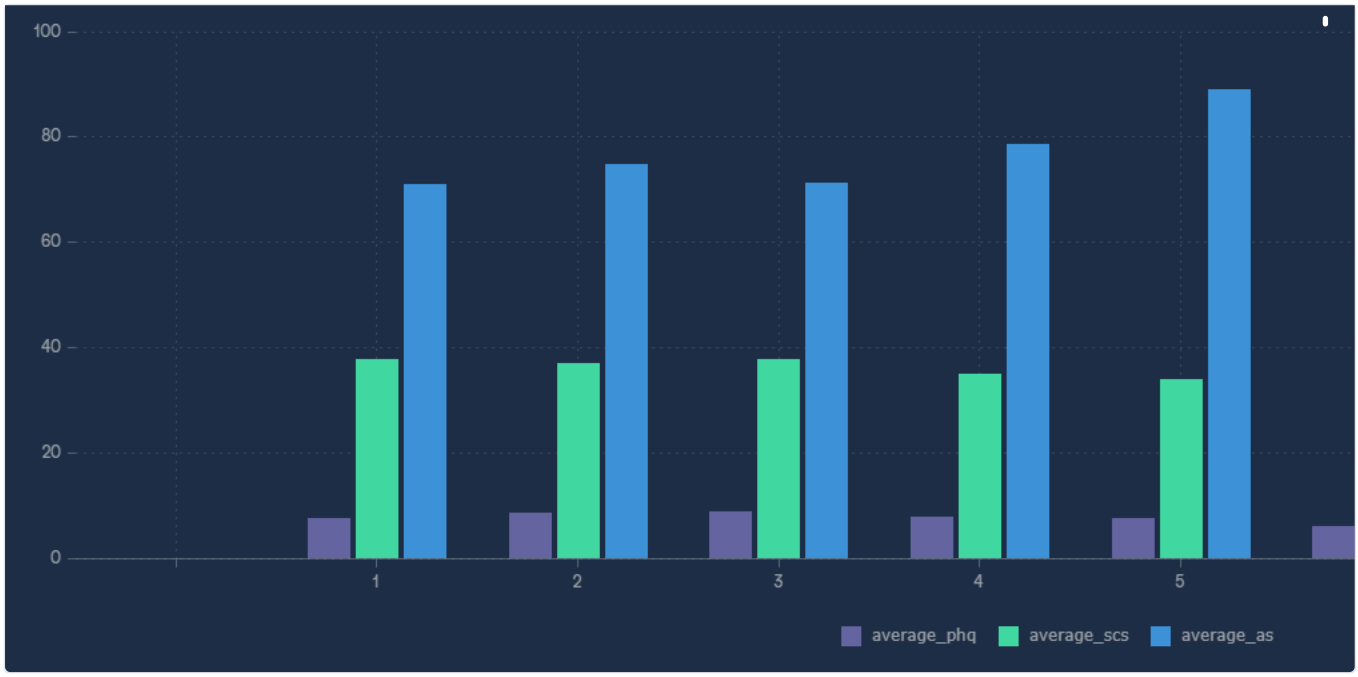
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40	Inter	JAP	Female	Under	21	3	10	Long	5	High	3	Ave
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218	Dom	JAP	Female	Under	20	2	3	Medium	5	High	2	Lov
219	Dom	JAP	Male	Under	21	3	2	Medium	5	High	3	Ave

Rows: 69 of 286 Expand

Answer with SQL

index	stay	count_int	average_phq	averc
0		0		
1	10	1	13	
2	8	1	10	
3	7	1	4	
4	6	3	6	
5	5	1	7.67	
6	4	14	7.96	
7	3	46	8.87	
8	2	39	8.58	
9	1	95	7.7	

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Key Insights:

- Critical Stress Threshold: Students in their 3rd year (stay = 3) exhibit a significant spike in acculturative stress (average_as = 71.35) backed by a robust sample size (n = 46).
- Data Quality & Outlier Management: Extreme metric variations observed in years 8 and 10 must be treated as statistical outliers due to a critically low sample size (n = 1). Decisions should not be weighted on these cohorts to prevent skewed budget allocations.